

# Kartikeya Gangwar

Undergraduate Researcher in Scientific Machine Learning, Geometric DL & Computational Mathematics

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## Education

University of Delhi

Delhi, India

B.Sc. (Honours) Mathematics (4-Year Undergraduate Research Degree)

2023 – 2027

- **Undergraduate Dissertation Supervisor: Prof. Vinay Kumar** (Department of Mathematics).
- **Core Specialization:** Real & Complex Analysis, Differential Equations, Symplectic Geometry, Numerical Methods, and Autodiff Systems developed through self-directed mathematical research alongside formal undergraduate coursework.

## Technical Expertise

**Scientific Machine Learning:** PINNs, Hamiltonian Neural Networks (HNNs), Symplectic Neural ODEs, Deep Surrogate Inversion, Fourier Positional Encodings, Multi-Task Mixture-of-Experts (MoE), Boundary-Encoded Neural Ansätze.

**Mathematical Foundations:** Symplectic Geometry, Canonical Differential 2-Forms, Extended Contact Phase Space, Nonlinear Parabolic/Elliptic PDEs (Navier–Stokes, Black–Scholes), Free-Boundary Obstacle Problems, Optimal Transport (Wasserstein Distance).

**Autograd & Deep Learning Systems:** PyTorch (`torch.func.vmap`, forward-mode autodiff, JVP/VJP, Hessian trace contractions), CUDA acceleration, GPU memory chunking, JAX, NumPy, SciPy, Matplotlib.

**Numerical Methods & HPC Infrastructure:** Finite Difference Schemes (ADI, Red-Black SOR), Symplectic RK4, L-BFGS (Strong-Wolfe), MCMC, C++, Lua, Linux/Bash, Git, LaTeX, Kaggle/Colab GPU Environments.

## Flagship Research Manuscripts & Preprints

Shape Inversion in Electrical Impedance Tomography

Under Review

*IEEE Transactions on Computational Imaging (TCI)*

[GitHub Code](#) | [Preprint DOI: 10.5281/zenodo.22096368](#)

- **Authors: Kartikeya Gangwar** (Sole Author / Lead Investigator).
- Parameterized closed interfaces via 64D periodic cubic B-splines; verified sensitivities  $\mathbf{J}_{\text{FEM}}$  against finite differences ( $< 3.2 \times 10^{-4}$ ).
- Introduced **Stochastic Directional JVP Supervision** on  $\mathbb{S}^{63}$ , reducing backprop tangent branches from 64 to 1 (342.8 MB peak VRAM).
- Delivered zero-online-FEM deterministic inversion with **56.8×** wall-clock speedup (35.6 ms vs 2.02 s/step) and 99.57% noise robustness.

CPA-SHNN: Symplectic Neural Networks for Celestial Mechanics

Working Paper

*Target: Astronomy & Computing / New Astronomy*

[GitHub Code](#)

- **Authors: Kartikeya Gangwar** (Lead Author), **Prof. Vinay Kumar** (Supervisor & Senior Author).
- Proved *Theorem 1* (Separable Symplectic Kinetic-Coriolis Decomposition,  $\nabla_{\mathbf{z}} \cdot \mathbf{f}_{\theta} \equiv 0$ ) and *Theorem 2* (Arnold Extended Contact Phase Space  $\mathbf{Z}_{\text{ext}} = (\mathbf{q}, t, \mathbf{p}, p_t)$ ,  $\mathcal{K}_{\theta} \equiv 0$ ), preserving energy to machine precision ( $\Delta \mathcal{H} \leq 10^{-4}\%$ ) across 6 chaotic systems.
- Multi-scale Fourier features achieved up to a **126.40×** error collapse on multi-body gravitational saddle singularities.

Operator Conditioning & False Convergence in Incompressible PINNs

Under Review

*Physics of Fluids (AIP Publishing)*

[GitHub Code](#) | [Preprint DOI: 10.5281/zenodo.22979276](#)

- **Authors: Kartikeya Gangwar** (Sole Author / Lead Investigator).
- Controlled regularized cavity study ( $Re = 1000$ ); showed  $\psi$ – $\omega$  exhibits **False Convergence** (67.3× residual drop while underpredicting secondary corner eddies by 14.3%, with 14.2% vs 2.8% global velocity  $L_2$  error).
- Proved continuous PINNs lack discrete Thom wall stencils; retaining pressure  $p$  in  $\psi$ – $p$  enforces implicit elliptic Helmholtz–Hodge projection, eliminating operator diffusion and capturing secondary vortex cores.

Extreme- $Re$  Navier–Stokes Framework & Unsteady Dynamics

Under Review

*Applied Mathematical Modelling (Elsevier)*

[GitHub Code](#) | [Preprint DOI: 10.5281/zenodo.18312938](#)

- **Authors: Kartikeya Gangwar** (Sole Author / Lead Investigator).
- Exact  $\mathcal{O}(N^2 \log N)$  DST-I spectral Poisson solver eliminating iterative residuals; zero artificial viscosity ( $\nu_{\text{art}} = 0$ ) to  $Re = 50,000$ .
- Computed steady continuation to  $Re = 100,000$  on  $1025^2$  grid (1.05M nodes); verified Batchelor’s core theorem and ASME V&V 20 GCI (0.48%).
- Resolved periodic vortex shedding ( $St = 0.6249$ ), modulated dynamics, and finite-window spectra to  $Re = 100,000$  on  $513^2$  grids.

## The Adaptive Subspace (AS) Optimization Trilogy

### Act I: Parameter Decoupling & Interface Transmission in PINNs

Under Review

*Computer Methods in Applied Mechanics and Engineering (CMAME, Elsevier)*

[GitHub Code](#) |

Preprint DOI: [10.5281/zenodo.22132799](#)

- **Authors:** Kartikeya Gangwar (First & Corresponding), Samarjeet Malik (Co-Author) [MS #CMAME-D-26-05120].
- Partitioned parameter manifold into orthogonal direct-sum subspaces ( $\Theta_0 \oplus \Theta_1, \mathcal{W}_0 \mathcal{W}_1^T = \mathbf{0}$ ) blended via  $C^2$  Quintic Hermite operator.
- Proved structural gradient orthogonality ( $\langle \nabla \mathcal{L}_{\text{if}}, \nabla \mathcal{L}_{\text{des}} \rangle \equiv 0$ ) on saturated domains; evaluated on 3D Linear Paul Ion Traps.

### Act II: Adaptive N-Subspace PINN (AS-PINN)

Preprint

Target: *Journal of Computational Physics (JCP) / SISC*

[GitHub Code](#) | [Preprint DOI: 10.5281/zenodo.22822521](#)

- **Authors:** Kartikeya Gangwar (Sole Author / Principal Investigator).
- Autonomous parameter-space AMR via vectorized Gram clash profiling (`torch.func.vmap`) with zero-disruption cleavage invariance.
- Validated across 9 canonical PDEs; achieved  $725.6\times$  lower loss on high-frequency Helmholtz ( $k = 4\pi$ ) over PCGrad and CAGrad.

### Act III: AS-ViT: Adaptive Subspace Vision Transformers

Preprint

Target: *IEEE TPAMI / CVPR*

[GitHub Code](#) | [Preprint DOI: 10.5281/zenodo.22974130](#)

- **Authors:** Kartikeya Gangwar (Sole Author / Principal Investigator).
- Resolved multi-task negative transfer in dense vision (segmentation, depth, normals) via feature-conditioned Partition of Unity (PoU) gating.
- Outperformed monolithic ViTs and fixed MoEs by  $+5.84\%$  multi-task gain ( $\Delta M$ ) across NYUv2, Cityscapes, and Taskonomy benchmarks.

## High-Dimensional Stochastics & Uncertainty Quantification

### High-Dimensional American Basket Options (Deep-EEP-PINN)

Working Paper

Target: *Quantitative Finance*

[GitHub Code](#)

- **Authors:** Kartikeya Gangwar (Lead Author / Principal Investigator).
- Priced correlated American basket options up to **d = 50 assets (1,225 correlations)** via Directional Autograd Hessian Trace Contraction in  $\mathcal{O}(d)$  linear complexity ( $< 3$  GB VRAM on Tesla P100). Validated against 100k-path Longstaff-Schwartz Monte Carlo (0.37% error).

### Bayesian PINN Fidelity & Parameter-Resolved Error Localization

Working Paper

Target: *Reliability Engineering & System Safety / Inverse Problems*

[GitHub Code](#)

- **Authors:** Kartikeya Gangwar (Lead Author / Principal Investigator).
- Formulated **Bayesian Fidelity Ratio (BFR)** based on 1-Wasserstein optimal transport distance normalized by paired MCMC stochastic noise floors.

## Scientific Software & Computational Infrastructure

### Extreme-Re Navier-Stokes FDM Suite & Benchmark Engine

Open-Source

*Python / SciPy / NumPy*

[GitHub Code](#) | [DOI: 10.5281/zenodo.18312938](#)

- **Lead Developer:** Kartikeya Gangwar. Reference-grade  $\psi$ - $\omega$  solver coupling exact DST-I Poisson acceleration with ADI continuation up to  $Re = 100,000$  on  $1025^2$  grids (1.05M nodes). Automated ASME V&V 20 GCI verification, directional hybrid differencing, and 30 verified flow-field datasets.

### BeamNG Research SDK (BRSDK v2.0.0)

Open-Source Framework

*Lua / Python / Systems & Sim2Real Robotics*

[GitHub Code](#) | [DOI: 10.5281/zenodo.21729606](#)

- **Architect & Lead Developer:** Kartikeya Gangwar. Open-source dual-runtime framework bridging the Sim2Real gap via **2,000 Hz in-engine soft-body telemetry** (tire slip energy, suspension velocities) with zero Lua heap allocations.
- Engineered columnar Python SDK on **Polars / Apache Arrow** with zero-copy **PyTorch (DLPack)** & **Minari** offline RL bindings, validated via strict Pydantic V2 sidecars and multi-platform CI.